

Design of a Multi-Agent Home Energy Gateway: Interpreting Household Plans and Configuring an Optimiser

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Abstract

In process.

Keywords

home energy management, energy gateway, multi-agent system, ReAct, large language model, model predictive control, plan interpretation, Design Science Research

Supervisor

Yueqiang Xu, PhD, Docent, Adjunct Professor

Paula Alavesa, PhD

Foreword

In process.

Abbreviations

BDEW	German Association of Energy and Water Industries (Bundesverband der Energie- und Wasserwirtschaft)
COP	Coefficient of Performance (of a heat pump)
DSR	Design Science Research
DWD	German Weather Service (Deutscher Wetterdienst)
EEG	Renewable Energy Sources Act (Erneuerbare-Energien-Gesetz)
EMC	Energy Management Controller
EMHASS	Energy Management for Home Assistant
EnWG	Energy Industry Act (Energiewirtschaftsgesetz)
EV	Electric Vehicle
HEMS	Home Energy Management System
LLM	Large Language Model
LP	Linear Programming
MILP	Mixed-Integer Linear Programming
MPC	Model Predictive Control
OCPP	Open Charge Point Protocol
OPSD	Open Power System Data
PV	Photovoltaic (rooftop solar panels)
ReAct	Reasoning and Acting
SLR	Systematic Literature Review
SoC	State of Charge (of a battery)
SQ	Sub-question

Declaration of Generative AI in the Thesis

In process.

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1. Introduction

1.1 Background and problem

A home energy gateway is the device in the meter cabinet or on the home network that reads the smart meter and controls the home's energy assets: the rooftop photovoltaic (PV) system and its inverter, a home battery, a heat pump, an electric vehicle (EV) charger and smart appliances. Its task is to decide, every few minutes, when each asset draws, stores or exports power, so that the household pays less, uses more of its own solar power and stays within the limits set by the grid operator. Figure 1.1 shows such a system.

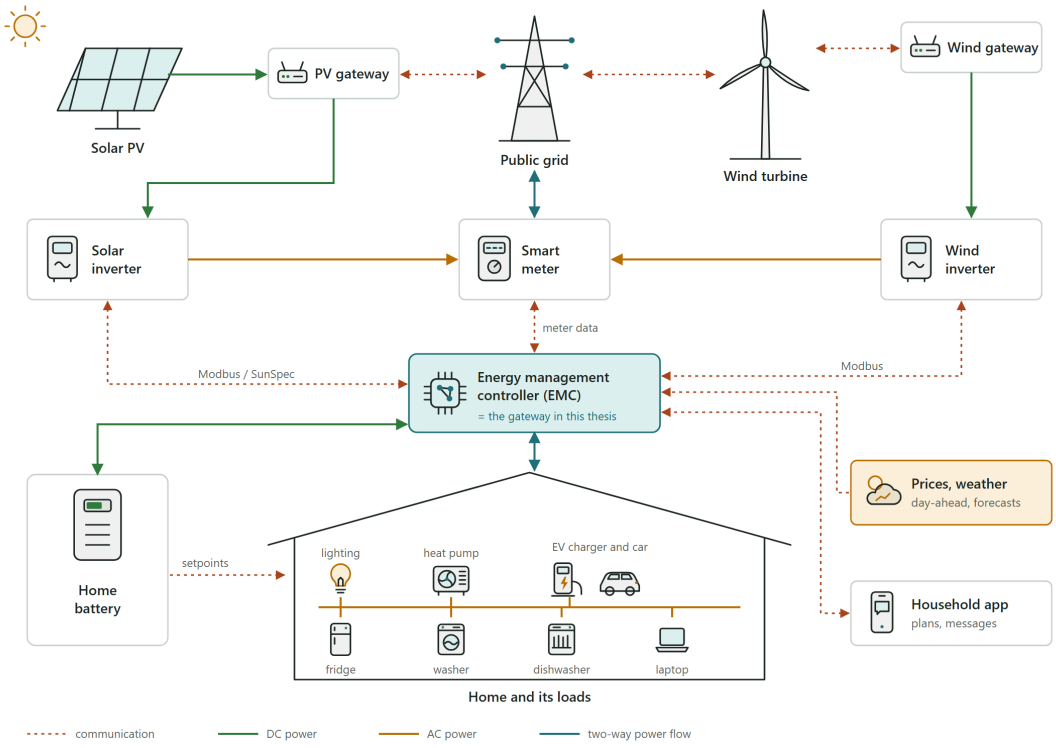


Figure 1.1. A home energy system and its controller. The energy management controller (EMC) is the gateway studied in this thesis.

Four problems limit what today's gateways achieve.

Devices do not share one language. Inverters, batteries, chargers and heat pumps from different makers communicate in different ways, many only through their manufacturer's cloud, and standards such as Matter and OpenADR are adopted slowly (Energy Systems Catapult, 2026).

Several controllers act on one device. An EV charger can receive commands from the car's app, the charger's app, a smart-charging service, a flexibility service and the energy

supplier (Energy Systems Catapult, 2026).

The grid operator sets limits. In Germany, heat pumps and wallboxes above 4.2 kW installed since 2024 must accept power reductions from the grid operator through the smart meter gateway and a control box (§14a of the Energy Industry Act, EnWG), so the household's plan has to fit inside an external limit (evcc, n.d.).

Planning by fixed rules leaves savings unused, and an optimiser needs an expert. Typical gateways follow rules such as "charge the car from PV surplus". Model predictive control (MPC), which plans ahead with price and weather forecasts, lowered the energy cost of a PV and battery home by 13.5 % against a rule-based strategy (Seal et al., 2019), and stochastic MPC performed best under realistic forecast errors (Blonsky et al., 2022). An MPC, however, needs exact parameters: when the car leaves and how full it must be, the comfort band of each period, the battery reserve and how far to trust the PV forecast. Households do not think in these parameters, settings become outdated when routines, seasons or tariffs change, and complicated systems slow adoption (Tuomela et al., 2021).

Language-model agents can bridge the gap between what a household says and what an optimiser needs. Recent work turns plain-language preferences into optimisation constraints (Sanguinetti et al., 2025; Michelon et al., 2025) or lets ReAct agents schedule appliances directly (El Makroum et al., 2026; Jonah et al., 2026). Direct schedulers come close to the optimum, but they also produced invented schedules and misstated deadlines, and their authors recommend a deterministic validator before any device is switched (Jonah et al., 2026). The author's earlier thesis reached the same conclusion for appliance scheduling: reliability and savings came from the solver and the deterministic checks, not from the language model or the number of agents (Le, 2026).

This thesis therefore gives the agents the work an optimiser cannot do: interpreting the household's plans and changes, configuring the optimiser from the house's own history, reacting to events and explaining the plan. The optimiser computes the schedule, and a deterministic checker and control layer switch the devices.

1.2 Research questions

The main research question is:

RQ: How can a language-model agent let a home energy gateway work from the household's plain-language plans without expert configuration, by interpreting the plans and configuring and explaining an optimiser, while keeping control reliable?

Four sub-questions (SQ) divide it:

SQ1 *Interpretation.* How accurately does the agent turn plain-language plans and their changes into constraints and a commit-or-ask decision, and how often are the household's requirements then met?

SQ2 *Configuration.* Can the agent configure the MPC from the house's history close to an expert-tuned MPC, and better than a deterministic auto-tuner that sees the same history?

SQ3 *Reliability and cost.* How often do the checker and the fallback intervene, what does the agent cost locally and in the cloud, and does splitting the work into four agents help compared with one or two?

SQ4 *Conditions.* Under which weather, price patterns, household loads and plan types does interpretation or configuration fail?

The thesis does not measure usability with users. Its claim is that the gateway works without expert configuration from plans written by people other than the author; a usability study is left for future work.

1.3 Research approach and scope

The thesis follows Design Science Research (DSR) (Hevner et al., 2004; Peffers et al., 2007): an artifact is designed, built, demonstrated and evaluated, with design iterations on an exploratory set and a registered confirmation on a held-out set (Chapter 3). The evaluation is limited to households: one home with PV, a battery, a heat pump and an EV charger is simulated with measured data, real weather and real prices (Section 3.3). Transferability to industrial energy management is discussed in Section 6.2.

1.4 Structure of the thesis

Chapter 2 reviews related work. Chapter 3 describes the research method, the data and the evaluation design. Chapter 4 presents the artifact. Chapter 5 reports the exploratory pilots; the registered evaluation is in process. Chapters 6 and 7 discuss and conclude.

2. Literature Review

The systematic literature review (SLR) defined in Section 3.2 is in process. This chapter summarises the studies already read, grouped by the parts of the design they inform.

2.1 Home energy management and optimisation

Home energy management systems (HEMS) schedule household loads, storage and generation against prices and comfort requirements; reviews classify the methods into rule-based control, mathematical optimisation, MPC and heuristics (Beaudin & Zareipour, 2015). MPC re-solves an optimisation over a receding horizon with updated forecasts. Against a rule-based strategy, a centralised MPC lowered the energy cost of a PV and battery home by 13.5 % (Seal et al., 2019). Under realistic uncertainty in weather, occupant behaviour and building parameters, stochastic MPC performed better than deterministic MPC and heuristics (Blonsky et al., 2022). Open-source tools bring optimisation into homes: EMHASS solves a day-ahead linear programme for Home Assistant installations (EMHASS contributors, 2026). The gap between such tools and households is configuration: the optimiser must be told the household's requirements and parameters, and the systems that require it are perceived as too complicated (Tuomela et al., 2021).

2.2 Language-model agents in home energy management

Language models are used in home energy management in two roles. *As an interface*, they translate what a user says into the inputs of an optimiser. Michelon et al. (2025) combine ReAct with few-shot prompting to extract the parameters a HEMS needs and report an average parameter retrieval accuracy of 88 %. Sanguinetti et al. (2025) map natural-language appliance preferences onto optimisation constraints for household batteries in an energy community. *As a scheduler*, agents plan loads themselves. El Makroum et al. (2026) coordinate several appliances with a ReAct orchestrator and specialist agents; Jonah et al. (2026) compare tool-calling agents on half-hourly prices and find that native function calling outperforms text-parsed actions and that low cost does not guarantee feasible behaviour. Raghavan and Giridhar (2025) pair a zero-shot agent with MPC, and multi-agent designs for home and building energy management are appearing (Jung, 2026; Yang et al., 2025; Liu et al., 2025). Zhang and Chen (2024) use a language model to make building control interpretable. The author's earlier thesis evaluated a hierarchical ReAct pipeline for appliance recommendations and found that the solver and the deterministic checks, not the agents, produced the savings and the validity (Le, 2026).

2.3 From natural language to optimisation models

Outside the energy domain, the NL4Opt competition defined the task of turning a word problem into an optimisation model (Ramamonjison et al., 2023). OptiMUS formulates and solves mixed-integer linear programmes from natural-language descriptions with a language-model agent that writes and debugs solver code (AhmadiTeshnizi et al., 2024), and Chain-of-Experts divides the formulation among cooperating expert agents (Xiao et al., 2024). These works generate the whole model; this thesis keeps a fixed, verified model and lets the agent set only its constraints and parameters.

2.4 Safety envelopes for unverified controllers

The Simplex architecture runs a high-performance but unverified controller inside an envelope of a simple, verified controller that takes over when the unverified one would leave the safe region (Sha, 2001). The deterministic checker and control layer of the gateway play the role of the verified envelope around the language-model agents.

2.5 Research gap

The studies above either let the language model produce schedules, which needs a validator, or extract parameters for a single request. None was found that (1) lets agents configure an MPC from a house's own history and compares them with a deterministic auto-tuner, (2) handles plans and changes during the day for PV, battery, heat pump and EV together, checked before execution, and (3) evaluates against a tuned rather than a default optimiser. Whether this gap holds is part of the SLR, which is in process.

3. Research Methodology

3.1 Design Science Research

The artifact is one home energy gateway with an agent layer of four agents and a deterministic harness (Chapter 4). Following Peffers et al. (2007), the work runs in three cycles:

1. *Cycle 1 (done):* exploratory prototypes: a digital twin of the house, a gateway pilot with planning agents, a plan-interpretation pilot and a Configurator pilot (Chapter 5).
2. *Cycle 2 (in process):* the tools and agents of the four-agent design, iterated on an exploratory set of house-days.
3. *Cycle 3 (in process):* a registered evaluation on a held-out set and a year simulation.

3.2 Systematic literature review

The review follows Kitchenham and Charters (2007) with backward and forward snowballing (Wohlin, 2014). Scopus and Web of Science are searched for peer-reviewed work and arXiv and Google Scholar for 2025 and 2026 preprints. The search string combines three blocks: home energy management terms ("home energy management", "residential demand response", "home energy gateway"), language-model terms ("large language model*", "LLM-based", "agentic", "ReAct") and, in a second run, optimisation terms ("model predictive control", "MILP", "scheduling"). Included are studies from 2022 to 2026 on behind-the-meter assets in which a language model generates constraints, tunes parameters, schedules loads or controls devices, and which evaluate the system. The search results are in process.

3.3 Data

Table 3.1 lists the data. The main house is site `DE_KN_residential4` of the OPSD Household data package (Open Power System Data, 2020): measured 15-minute energy data from homes in Konstanz, Germany, with PV generation, grid import and export, a heat pump, an EV and appliances. In 2016 the house's PV produced about 11.0 MWh, the heat pump used 2.6 MWh and the EV 1.0 MWh. Weather, prices and heat pump efficiency come from the same place and dates.

The import price is a counterfactual dynamic tariff built from the 2016 components: (spot price + supplier margin + 17.94 ct/kWh of fees and levies) $\times$ 1.19 VAT. A PV model fitted to DWD global radiation reproduces the house's PV with a correlation of 0.96 hourly and

Table 3.1 Data used in the evaluation.

Data	Content and source	Use
OPSD Household data	15-minute PV, grid, heat pump, EV and appliance data of homes in Konstanz, 2014–2019 (Open Power System Data, 2020)	main house; more houses for SQ2
DWD station 02712	hourly air temperature and global radiation, Konstanz (Deutscher Wetterdienst, n.d.)	weather classes, forecasts
Day-ahead prices	German spot prices, 2015–2019 (Fraunhofer ISE, n.d.)	import and export price
BDEW price analysis	2016 household price components, 28.80 ct/kWh (BDEW, 2023)	dynamic tariff
EEG feed-in tariff	12.31 ct/kWh for rooftop PV up to 10 kWp (Bundesnetzagentur, n.d.)	export price
When2Heat	heat pump COP by hour (Ruhnau et al., 2019; Ruhnau & Muessel, 2023)	heat demand from heat pump electricity
Norwegian EV sessions	35,377 residential charging sessions (Sørensen et al., 2024)	optional: departure times
Plan benchmark	plain-language plans with answer keys (pilot set in Appendix A)	SQ1 ground truth

0.98 daily.

3.4 Evaluation design

Benchmark. The unit is a house-day of 96 fifteen-minute slots with a plain-language plan and, where the design calls for it, a change during the day. Factors are the plan type (structured, plain language, plain language with a change, ambiguous, conflicting or infeasible), weather, price pattern, load and house. Plans for the held-out set are written by three to five people other than the author, keyed before any run and checked by a second person.

Systems. All systems use the same house-days, forecasts, plans and control layer: the recorded operation; rules; EMHASS; default MPC; an expert-tuned MPC (settings chosen once on exploratory weeks and frozen); a deterministic auto-tuner (weekly grid search over past weeks); a language model without tools; the four-agent gateway with a cloud and a local model; ablations with one and two agents; and perfect foresight as the upper bound.

Metrics. Constraint precision and recall against the answer key, clarification F1 and decision accuracy (SQ1); household requirements met; adjusted cost, which subtracts the value of battery charge and stored heat left at midnight; the share of the gap between rules and perfect foresight that a system closes; rejected plans, fallbacks and failed calls; tokens, cost, mean and 95th-percentile latency and energy per event.

Hypotheses (draft, to be registered). H1: constraint precision and recall of at least 0.90 and a clarification F1 of at least 0.85; H2: household requirements met on at least 95 % of feasible house-days; H3: plain-language plans cost no more than the same plans given as exact parameters to the tuned MPC, within 5 % of the rules-to-optimum gap; H4a and H4b: the agent-configured MPC closes a registered share of the gap between default and tuned MPC and is non-inferior to the auto-tuner; H5: the best local model is within 0.10 of the cloud model's recall; H6: the yearly saving exceeds the gateway's own cost with cloud inference and with a local inference box. The analysis uses cluster bootstrap intervals over plans, exact binomial and paired Wilcoxon tests, and Holm correction within SQ1 and within SQ2.

Validity. The main threats and their handling are author-written plans (other writers, second-person check), tuning on evaluation days (tuning only on exploratory weeks), a harness that passes H2 by construction (scoring against the answer key, with grid and battery limits reported separately as safety checks), mixed data sources (same place and dates, the dynamic tariff stated as counterfactual) and simulation only (a field study as future work).

4. System Design and Implementation

4.1 Requirements

From the problem (Section 1.1) follow five requirements: (R1) the household states its plans in plain language; (R2) the gateway is configured without an expert; (R3) no language model computes a schedule or switches a device; (R4) every plan passes a deterministic check before it takes effect; (R5) the house stays within its limits when an agent, a model or the network fails.

4.2 Architecture

The gateway has three layers (Figure 4.1). The *agent layer* holds four agents, each started by one kind of event. The *harness* holds the tools, a tuned MPC that replans every hour on its own, a checker and a history store. The *control layer* runs every 15 minutes, holds the planned grid exchange so that the battery absorbs forecast errors, keeps comfort, the EV deadline and the battery limits, and falls back to rules. The harness follows the Simplex pattern (Sha, 2001): the agents are unverified controllers inside a verified envelope.

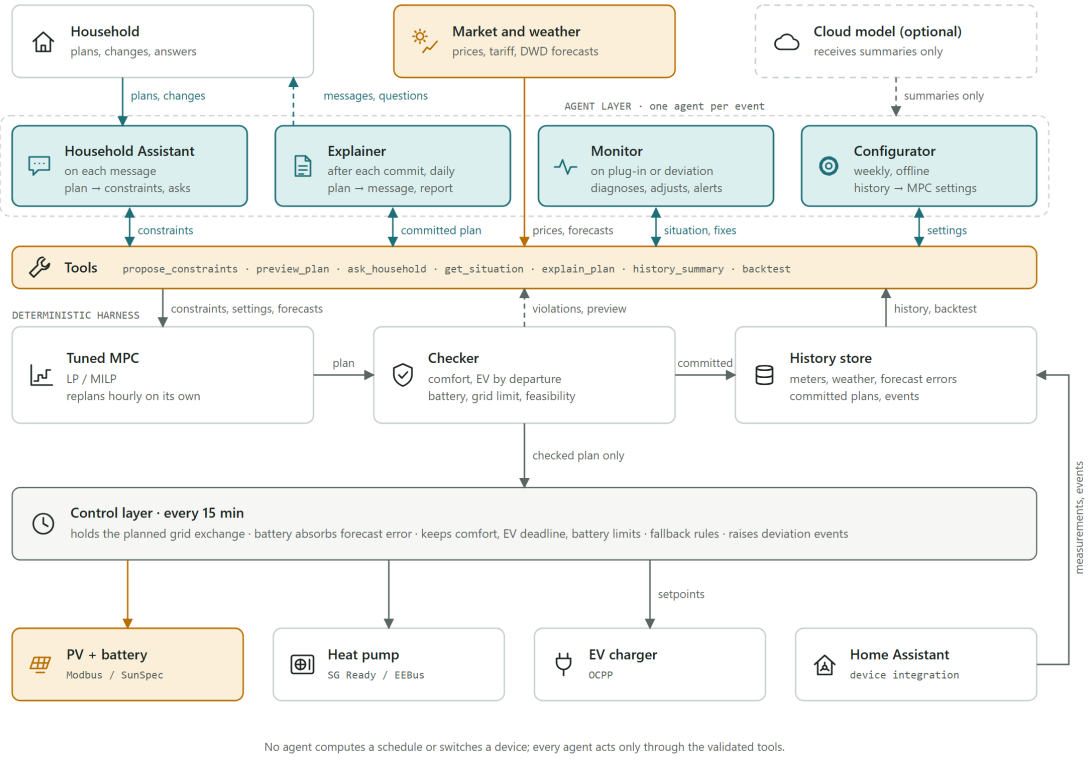


Figure 4.1. Architecture of the gateway: four agents act only through validated tools; the tuned MPC plans and the checker passes only valid plans to the control layer.

4.3 Four agents, one per event

Table 4.1 lists the agents. The harness routes each event to one agent by its type; no agent calls another, and the agents share state only through the harness. Splitting by event keeps each prompt small and lets each output be checked on its own. The earlier thesis found that splitting *one* task between agents added cost without reliability (Le, 2026); here the agents do different tasks at different times, and the agent count is tested as part of SQ3.

Table 4.1 The four agents.

Agent	Started by	Job
Household Assistant	a message from the household	turns the plan or change into constraints, previews them, commits or asks
Explainer	each commit; once a day	writes the message and the daily report from the committed plan's figures
Monitor	an EV plug-in; a deviation raised by the control layer	finds the cause, adjusts constraints within what the household asked, or alerts
Configurator	once a week	proposes MPC settings from the house's history and backtests them

Every agent follows the same pattern (Figure 4.2): it uses its tools in a short loop and ends at a deterministic gate. The checker gates the Assistant and the Monitor, a figure check gates the Explainer, and a backtest gates the Configurator. Only a passed gate changes the plan, the message or the settings.

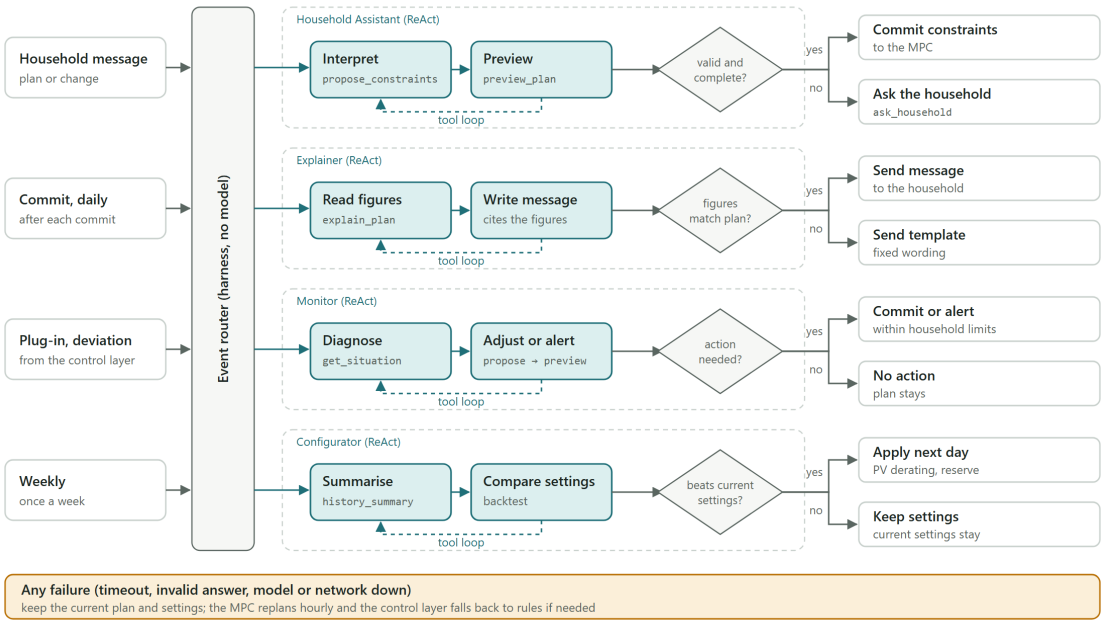


Figure 4.2. How each agent decides.

4.4 Tools

The Assistant reads the plan (`read_plan`), the situation (`get_situation`) and the forecast (`get_forecast`), proposes constraints (`propose_constraints`: EV energy and departure, comfort bands by time, away periods and appliance windows, validated against ranges), previews the plan with the MPC and the checker (`preview_plan`), and asks the household only for ambiguity, conflict or infeasibility (`ask_household`). The Explainer words pre-computed figures (`explain_plan`); every figure it cites is compared with the plan. The Configurator reads weekly statistics (`history_summary`) and compares candidate settings on past weeks (`backtest`). Feasibility, for example more energy than the charger can deliver before departure, is computed by the checker, not by the language model. Figure 4.3 shows one day at runtime.

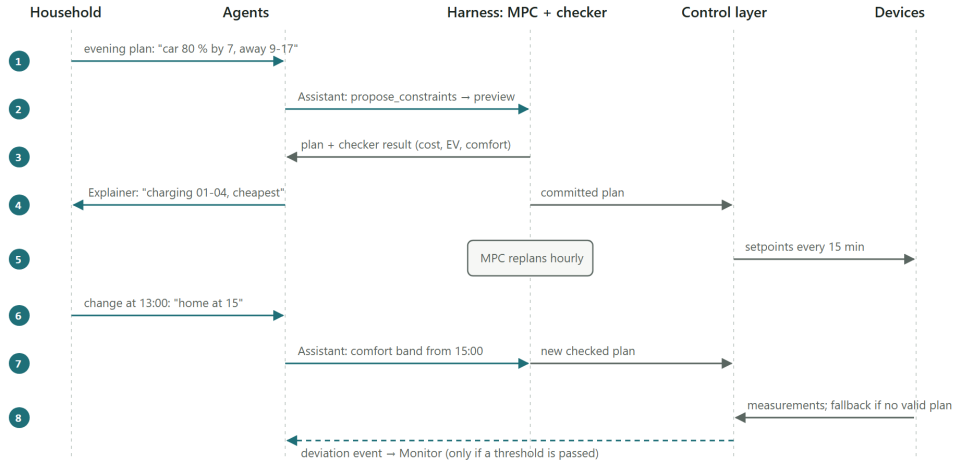


Figure 4.3. One day at runtime. Between agent calls the MPC replans every hour and the control layer runs every 15 minutes without a language model.

4.5 Implementation

The prototypes are written in Python. The digital twin replays the OPSD house with a virtual 10 kWh / 5 kW battery; the MPC is a linear programme solved with PuLP and CBC from the measured state to midnight, and energy left at midnight is valued in the same way in the optimiser and in the score. The agents use LangChain and LangGraph ReAct agents with structured outputs and a recovery step that returns invalid answers with the validation errors. Models are GPT-6 Luna through the OpenAI API and local models through Ollama. The implementation of the full four-agent design is in process.

5. Evaluation and Results

All results in this chapter are exploratory: one house, one run per configuration and draft answer keys. They guided the design and are not tests of the hypotheses.

5.1 Gateway pilot: agents as planners

The first pilot let a ReAct agent plan the gateway at four checkpoints a day (00, 06, 12 and 18 h and each EV plug-in), with an MPC tool whose settings it chose. Twelve days of 2016, one per month, were run with German prices and weather-driven forecasts. Table 5.1 shows the share of the cost gap between rules and perfect foresight that each strategy closed.

Table 5.1 Gateway pilot, 12 days of 2016: adjusted cost (negative values are net income from feed-in) and share of the rules-to-optimum gap closed.

Strategy	Adjusted cost (EUR)	Gap closed
Rules	−4.79	0 %
Day-ahead plan	−5.50	19 %
Default MPC at checkpoints	−5.65	23 %
Default MPC hourly	−5.97	31 %
Agent, GPT-6 Luna	−6.13	36 %
Agent, gpt-oss-20b (local)	−6.25	39 %
MPC with PV derating 0.9, at checkpoints / hourly		46 % / 51 %
Perfect foresight	−8.54	100 %

The agents beat the default MPC because they planned with less of the PV forecast; one fixed setting did this better. GPT-6 Luna made 63 checks with no errors for USD 0.025 in total; gpt-oss-20b made 63 checks with three malformed calls caught by the fallback and took about 152 s per day. The derating of 0.9 was chosen on the same 12 days, a weakness the later pilot exposed (Section 5.3). The pilot changed the design: the agent no longer plans, the MPC replans hourly without it, and the control layer holds the planned grid exchange instead of the planned battery power.

5.2 Plan-interpretation pilot

The second pilot tested SQ1 on 22 edge-case plans for the main house (Appendix A): six simple plans, five changes during the day, six ambiguous and five conflicting or infeasible requests, each bound to a date of 2016 and the car's charge. The model returned the plan as a structured `HouseholdPlan` with a commit-or-ask decision, through function calling

with up to two validation retries. Items matched the answer key within 15 minutes, five percentage points of EV charge or 1 kWh.

Table 5.2 Plan-interpretation pilot, 22 plans.

Model	Precision	Recall	Clarif. F1	Decision	s per plan
GPT-6 Luna (cloud)	1.00	1.00	0.95	0.95	2.7
gpt-oss-20b (local)	0.63	0.44	0.75	0.59	116
Qwen3 8B (local)	0.96	1.00	0.67	0.68	71
Gemma 4 E4B (local)	1.00	0.91	0.78	0.82	7.7
Qwen3 30B-A3B (local, CPU)	1.00	0.96	0.84	0.86	150

GPT-6 Luna made one error: for "charge the car to 90 % before I head out" it assumed a departure time and committed instead of asking. gpt-oss-20b often left numbers such as the EV target out and asked questions that simple plans did not need. Three newer open-weight models ran on the same laptop (RTX 4070, 8 GB), with output capped at 8,000 tokens per call after Qwen3 8B looped in its reasoning phase on one plan. All three extracted items almost as well as the cloud model; they differed in when they asked. Qwen3 8B committed on almost half of the plans that needed a question, Gemma 4 E4B missed most conflicts but answered in under 8 seconds, and Qwen3 30B-A3B came closest to the cloud model but had to run on the CPU and took two and a half minutes per plan. The cloud model is close to the ceiling on these cases, which is why the held-out plans are written by other people. No local model yet meets H5 on both accuracy and latency; Gemma 4 E4B and Qwen3 30B-A3B are the candidates for the exploratory phase.

5.3 Configurator pilot

The third pilot tested the riskiest part of SQ2: whether a language-model Configurator sets the MPC better than a deterministic auto-tuner with the same history. Over eight weeks (7 March to 1 May 2016), before each week the Configurator (GPT-6 Luna) read two weeks of history, backtested candidate settings and proposed a PV derating and a battery reserve; the harness applied them only if their backtest was not worse than the current settings. The auto-tuner chose the best of ten grid points on the same two weeks. The expert setting was tuned once on four September weeks (derating 0.7).

Table 5.3 Configurator pilot, eight weeks of 2016: saving against the default MPC.

System	Saving vs default (EUR)	Rules-to-optimum gap closed
Default MPC	0.00	36 %
Auto-tuner (two weeks of history)	0.10	36 %
Configurator agent, GPT-6 Luna	2.82	48 %
Expert setting (tuned on September)	2.83	48 %
Perfect foresight	14.45	100 %

The agent matched the expert setting; the auto-tuner, which switched between deratings of 0.6 and 0.9, overfitted two weeks of history, while the agent kept stable settings between 0.75 and 0.9. The agent cost USD 0.003 in total and took about 16 s and five tool calls a week. In a one-week test, a setting tuned on another single week did worse than the default, so the tuned reference must always be chosen on separate weeks.

5.4 Registered evaluation

In process.

6. Discussion

6.1 Answers to the research questions

In process.

6.2 Transferability to industrial energy management

The design is not specific to households. Table 6.1 maps its parts to an industrial site, where the household's plan becomes a shift plan, a production order, a maintenance window or an operator's note.

Table 6.1 Gateway components and their industrial counterparts.

Gateway (home)	Industrial site
Household plan and changes	shift plans, production orders, maintenance windows, operator notes
Household Assistant	operator assistant: orders and notes into constraints, questions when unclear
Configurator	learns flexible windows, peak limits and forecast caution from plant meter history
Monitor	reacts to line stops, late orders and peak-limit warnings
Explainer	daily energy report for the energy manager
Tuned MPC	production-aware scheduling model with job order and minimum run times
Checker and control layer	the same role, with process safety and quality limits
Home Assistant, Modbus, OCPP	SCADA and MES systems over OPC UA or Modbus

What changes is the size and kind of the optimisation (job dependencies and minimum run times), the cost structure (peak demand charges and contracts often matter more than spot prices), security and confidentiality of production data, which favour a local model, and the user, an operator or energy manager who needs a record of each decision. Exploratory prototypes on an electronics factory week from the HIPE dataset (Bischof et al., 2018) and on a small business in the OPSD data (Open Power System Data, 2020) found that the agents and a simple smart timer reached the optimum on every day, so, as in homes, the open problems are interpretation and configuration rather than optimisation. Real deadlines and job order were unknown in both datasets, so these findings are not results; testing plans written for a plant is future work.

6.3 Limitations

In process.

6.4 Future research

In process.

7. Conclusions

In process.

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Appendix A. Pilot plans and answer-key rules

The 22 plans of the plan-interpretation pilot are in the file `prototypes/plan_pilot/plans.json` of the thesis repository. Cases S1 to S6 are simple, C1 to C5 change the plan during the day, A1 to A6 are ambiguous and X1 to X5 are conflicting or infeasible. Twenty were adapted from an external review and two restored from the first pilot. The answer keys are drafts and are checked before any result is used. They follow five rules:

1. A request that the charger or the house cannot meet is an infeasible plan, and the expected decision is to keep the current plan and ask.
2. After a change during the day, the key holds only the requirements still in force; a period already running is scored from the time of the change.
3. Candidate constraints and the commit decision are scored separately: on an ambiguous plan the agent must not commit, but the clear parts still count towards precision and recall.
4. An away period is recorded as such; a comfort band is expected only when the household names a temperature.
5. Times are absolute dates and times, and only appliances the house has are used.

Appendix B. Tool and output schemas

In process.